

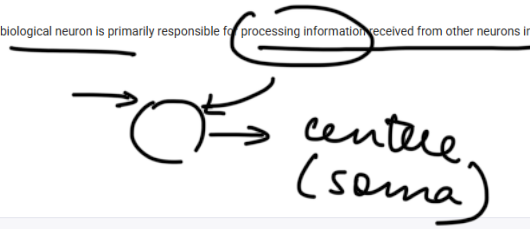
Week 1 PA GA – DL

PA

1) Which of the following components of a biological neuron is primarily responsible for processing information received from other neurons in the brain?

1 point

- ☐ Dendrites
- ☒ Soma
- ☐ Axon
- ☐ Synapses



2) Suppose that the MP neuron model is used to predict whether a particular user will buy a smartphone ($y = 1$) or not ($y = 0$). Suppose further that the user looks for a set of specifications listed below

1 point

Specification	Yes(1), No(0)
Front Camera	1
Screen > 6 inch	1
Price < 40K	0
5G Support	0
Branded	1
Weight < 170g	0
not made in India	0

The user buys the phone at least if any two of his/her specifications are present, and rejects the phone (despite the other specifications) if it is not made in India.

What is the suitable threshold (θ) value so that the user buys the phone?

- ☐ 1
- ☐ 0
- ☒ 2
- ☐ 3
- ☐ 4

$$\theta = 2$$

inhibitory

3) In question 2, Which of the specifications listed in the table is an inhibitory input to the MP neuron?

1 point

- ☐ Price
- ☐ 5G support
- ☒ not made in India
- ☐ None of those

4) The statement that all Boolean functions can't be implemented only using excitatory inputs is

1 point

- ☐ True for non-linearly separable Boolean functions
- ☒ True for some Boolean functions independent of whether they are linearly separable or not
- ☐ False, all Boolean functions can be implemented without using inhibitory inputs

5) Suppose that we represent inputs $x \in \{0, 1\}^6$ of a Boolean function as vertices of a hypercube in 6 dimensional space. Then the hyperplane defined by $x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 0$ only touches which of the following input?

1 point

- ☐ (0,0,0,1,0)
- ☐ (0,0,0,0,1)
- ☐ (0,1,1,0,0,1)
- ☒ (0,0,0,0,0)

6) We know that a Boolean function maps $\{0, 1\}^n \rightarrow \{0, 1\}$. How many linearly separable functions are there when $n = 1$?

1 point

- ☐ 0
- ☐ 1
- ☐ 2
- ☒ 4

$$2^n = 2^2 = 4$$

7) We know from the lecture that the decision boundary learned by the perceptron is a line in $\mathbb{R}^2$. We also observed that it divides the entire space of $\mathbb{R}^2$ into two regions, suppose that the input vector $x \in \mathbb{R}^5$, then the perceptron decision boundary will divide the whole $\mathbb{R}^5$ space into how many regions?

1 point

- ☒ 2
- ☐ 3
- ☐ 5
- ☐ 4
- ☐ It depends on whether the data points are linearly separable or not

still the pt. remains the same, we get 2 halfs of the same region.

- 8) Consider the following table, where x_1 and x_2 are features (packed into a single vector $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$) and y is a label,

x_1	x_2	y
0	0	0
0	1	1
1	0	1
1	1	1

1 point

Suppose that the perceptron model is used to classify the data points. Suppose further that the weights $\mathbf{w}$ are initialized to $\mathbf{w} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. The following rule is used for classification,

$$y = \begin{cases} 1, & \text{if } \mathbf{w}^T \mathbf{x} > 0 \\ 0, & \text{if } \mathbf{w}^T \mathbf{x} \leq 0 \end{cases}$$

The perceptron learning algorithm is used to update the weight vector $\mathbf{w}$. Then, how many times the weight vector $\mathbf{w}$ will get updated during the entire training process?

- ☒ 0
- ☐ 1
- ☐ 2
- ☐ 3
- ☐ Not possible to decide

at not any place the wt. vector actually gets updated

- 9) Consider the following table, where x_1 and x_2 are features and y is a label

x_1	x_2	y
0	0	1
0	1	1
1	0	1
1	1	0

1 point

Assume that the elements in $\mathbf{w}$ are initialized to zero and perceptron learning algorithm is used to update the weights $\mathbf{w}$. If the learning algorithm runs for long enough iterations, then

- ☒ The algorithm converges (i.e., no further weight updates) after some iterations
- ☐ The algorithm never converge
- ☒ The classification error becomes zero eventually
- ☐ The classification error remains greater than zero

- 10) The statement that "In perceptron model, the weight vector $\mathbf{w}$ is always orthogonal to the linear decision boundary" is

1 point

- ☒ True
- ☐ False

$$\mathbf{w}^T \mathbf{x} = 0$$

GA

- 1) Which of the following threshold values of MP neuron implements AND Boolean function? Assume that the number of inputs to the neuron is 10 and the neuron does not have any inhibitory inputs.

1 point

- ☐ 1
- ☒ 10
- ☐ 11
- ☐ 12

for AND, all the inputs must be true to fire off the neuron.
so, the no. of neurons will be equal to the no. of inputs.

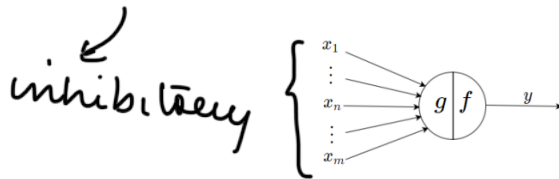
total m data pts.

n +ve data pts. ←

$(m-n)$ -ve data pts.

2) The diagram below shows an MP neuron.

1 point



Suppose that we define the function g as follows,

$$g(x_1, x_2, \dots, x_m) = \left(\sum_{i=1}^n x_i \right) \prod_{j=n+1}^m (1 - x_j).$$

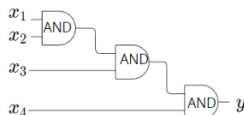
The output from the neuron is given by,

$$f(g(\mathbf{x})) = \begin{cases} 1, & \text{if } g(\mathbf{x}) \geq \theta \\ 0, & \text{if } g(\mathbf{x}) < \theta \end{cases}$$

How many inputs are inhibitory in the neuron? Assume θ is positive valued.

- ☐ 0
- ☐ m
- ☐ n
- ☒ $m-n$
- ☐ $m-n+1$

3) The diagram below shows an implementation of a Boolean function using three logical AND gates. What is the minimum number of MP neurons required to implement the same Boolean function? Assume that all inputs are excitatory. 1 point



- ☐ 3
- ☐ 2
- ☒ 1
- ☐ 4

fire if $\sum w_i x_i \geq 0$
we can keep $\theta = 0$

4) We know that a Boolean function maps $\{0, 1\}^n \rightarrow \{0, 1\}$. How many Boolean functions are there when $n = 0$? Assume that the function allows an empty element as an input. 1 point

- ☐ 0
- ☐ 1
- ☒ 2
- ☐ 4

2 → 2

Yes, the answer is correct.

Score: 1

Accepted Answers:

2

5) An input to the perceptron model is, $\mathbf{x} = [x_0, x_1, \dots, x_n]^T \in \mathbb{R}^{n+1}$ and the weight vector is $\mathbf{w} = [w_0, w_1, \dots, w_n]^T$. The output from the perceptron is given by 1 point

$$f(g(\mathbf{x})) = \begin{cases} 1, & \text{if } g(\mathbf{x}) \geq 0 \\ 0, & \text{if } g(\mathbf{x}) < 0 \end{cases}$$

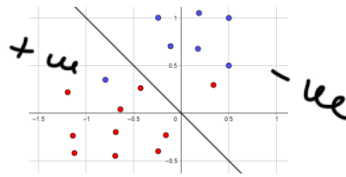
where $g(\mathbf{x}) = \mathbf{w}^T \mathbf{x}$, then the decision boundary (i.e. the hyper-plane that separates the data points) passes through the origin in?

- ☐ n^{th} dimension
- ☒ $n+1^{\text{th}}$ dimension
- ☐ $n-1^{\text{th}}$ dimension
- ☐ It is not possible to decide whether it passes through the origin in any of the dimension

$g(\mathbf{x}) = \mathbf{w}^T \mathbf{x} = 0$
hyperplane
this hyperplane
lies $(n+1)^{\text{th}}$ dim.

6) In the diagram shown below, the blue data points belong to positive class ($\mathbf{w}^T \mathbf{x} \geq 0$) and red data points belong to negative class ($\mathbf{w}^T \mathbf{x} < 0$). The number of misclassified data points according to the decision line as shown in the figure is?

1 point



☐ 0

☐ 1

☒ 2

insufficient information

most likely but again because it has not been told which is actually the -ve half space

7) The table below shows the truth table for NAND gate (a Boolean function). Suppose that the perceptron model is to be used for learning the decision line such that it separates all the data points with zero classification error

1 point

x_1	x_2	y
0	0	1
1	0	1
0	1	1
1	1	0

The value of the bias term (w_0) should strictly be

☒ Positive

☐ Non-negative

☐ negative

☐ Depends on the other weight (w_1, w_2) values

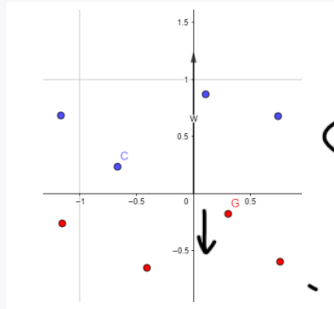
$$y = \begin{cases} 1 & w_1 x_1 + w_2 x_2 + w_0 \geq 0 \\ 0 & w_1 x_1 + w_2 x_2 + w_0 < 0 \end{cases}$$

$$w_0 \geq 0$$

Common data for question 8,9 and 10

In the figure shown below, the blue points belong to class 1 (positive class) and the red points belong to class 0 (negative class). Suppose that we use a perceptron model, with the weight vector w as shown in the figure, to separate these data points. We define the point belongs to class 1 if $w^T x \geq 0$ else it belongs to class 0.

$$w^T x \geq 0$$



+ve
①

-ve
②

8) The point G and C will be classified as?

1 point

Note: the notation $(G, 0)$ denotes the point G will be classified as class-0 and $(C, 1)$ denotes the point C will be classified as class-1

- ☐ $(G, 0), (C, 0)$
- ☐ $(G, 1), (C, 0)$
- ☒ $(G, 0), (C, 1)$
- ☐ $(G, 1), (C, 1)$

$$\begin{aligned} (w^T x)_G &< 0 \\ (w^T x)_C &\geq 0 \end{aligned} \Rightarrow \begin{aligned} &0 \\ &1 \end{aligned} \Rightarrow \begin{aligned} &(G, 0) \\ &(C, 1) \end{aligned}$$

9) The statement that "there exists more than one decision lines that could separate these data points with zero error" is,

1 point

- ☒ True
- ☐ False

Yes, the answer is correct.

Score: 1

Accepted Answers:

True

there can be many such lines actually

10) Suppose that we multiply the weight vector w by -1 . Then the same points G and C will be classified as?

1 point

- ☐ $(G, 0), (C, 0)$
- ☒ $(G, 1), (C, 0)$
- ☐ $(G, 0), (C, 1)$
- ☐ $(G, 1), (C, 1)$

just opp. to what happened in Q8
 $\downarrow w$
 $G \rightarrow 1$
 $C \rightarrow 0$